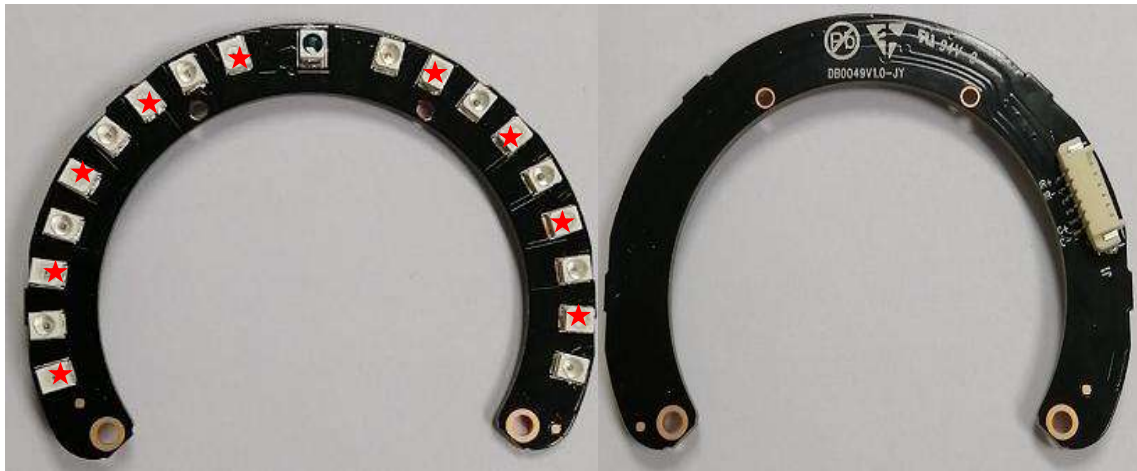


1. Product Images

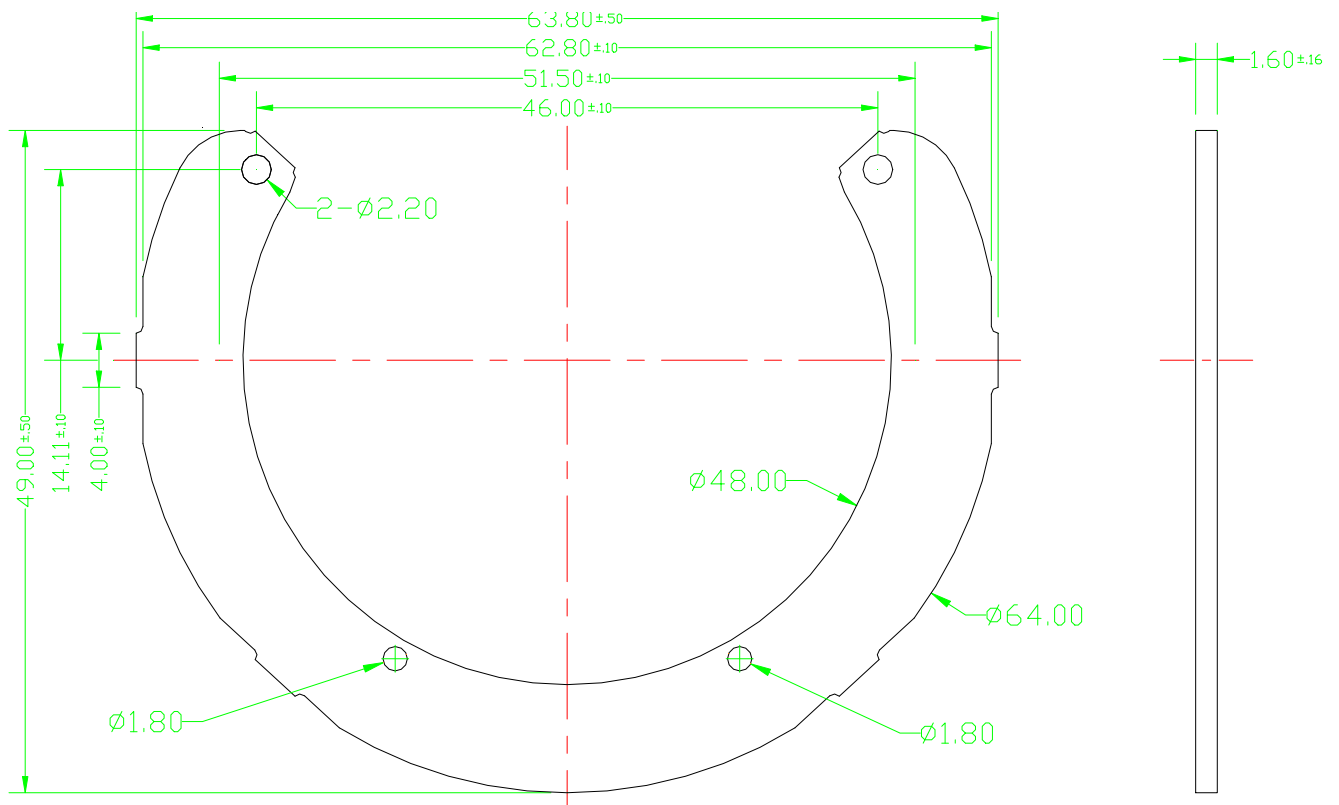


Front

Back

Note: The marked  position is 120 degrees (9 pieces), the remaining positions are 60 degrees (9 pieces)

2. Assembly dimensions



3. Product Overview

Product Model	DB0049V1.0-JY	PCB Material	FR4
Lamp Board Length and Width	(63.8*49)±0.50mm	LED Quantity	60° (9 PCS) 120° (9 PCS)
Lamp Board Inner Diameter	φ48±0.20mm	Colloid Color	Colorless Transparent
PCB Thickness	1.6±0.16mm	Colloid Appearance	Square
LED Bead Height	2.02/1.05±0.15mm	Chip Size	2835 14mil 60/120°
Overall Board Thickness	7.36±0.50mm	Emission Wavelength	850nm

4. Technical Parameters

Test Item	Required Range	Unit	Test Condition	Test Tools
Rated Current	/	mA	Room Temperature	Multimeter, Digital Power Supply
Rated Voltage	/	V	Room Temperature	
Illuminance Range On	/	Lux	Dark Box, Room Temperature	Lux Meter
Illuminance Range Off	/	Lux	Dark Box, Room Temperature	Lux Meter
IR-CUT Output Voltage	/	V		
IR-CUT output polarity	/	V		
Lamp board operating temperature	-20~+85	°C	after assembly	
Lamp board storage temperature	-40~+85	°C		

5. LED optoelectronic parameters (TA=25°C)

Test item	Specification	Value			Unit
		Min	Typ	Max	
Forward current	IF	--	--	70/80	mA
Forward voltage	VF	1.4	--	1.8	V
Reverse current (VR=5V)	IR			10	uA
Power consumption	PD	--	--	120/200	mW
Peak Wavelength	λp	840	--	860	nm
Radiant Intensity	Ie	--	24/15	--	mW/sr
Angle	θ(50%)	--	60/120	--	Deg
Operating Temperature	Topr	-20	--	+75	°C
Storage Temperature	Tstg	-30	--	+80	°C
Forward Pulse Current (1/8 duty f=1kHz)	I fp	--	--	1000/1500	mA
Electrostatic (HBM)	ESD	--	--	2000	V
Reflow soldering temperature	Tsol	--	--	230	°C
Luminous decay	<1%@3000 hours 85°C				

<6%@10000 hours 85°C
<20%@30000 hours, 85°C

6: LED Curve Chart

Fig 1. Forward Current vs. Forward Voltage

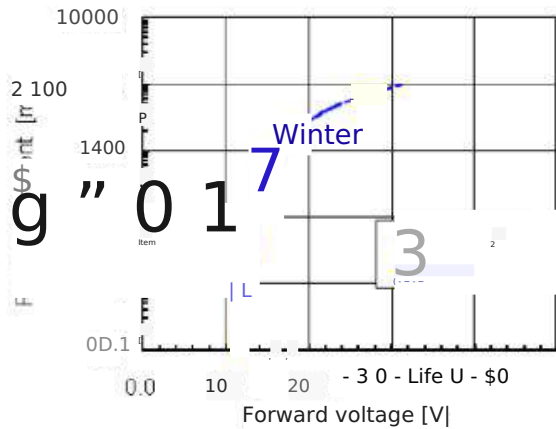


Fig 2. Forward Current vs. Forward Voltage

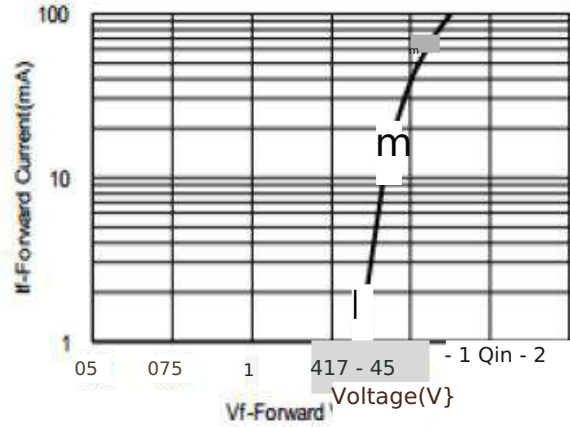


Fig 3. Forward Voltage(@100mA) vs. Ambient Temperature

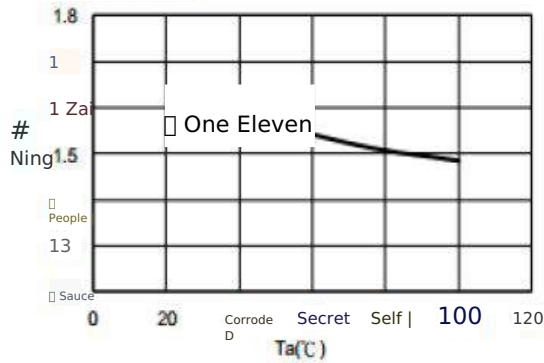


Fig 4. Forward Voltage(@100mA) vs. Ambient Temperature

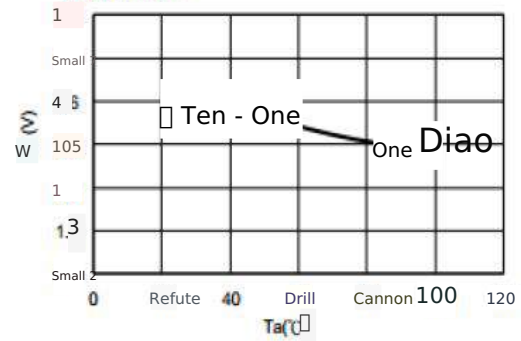


Fig.6- Maximum Driving Forward DC Current vs Ambient Temperature (Derating based on $T_{j\max}=110^{\circ}\text{C}$)

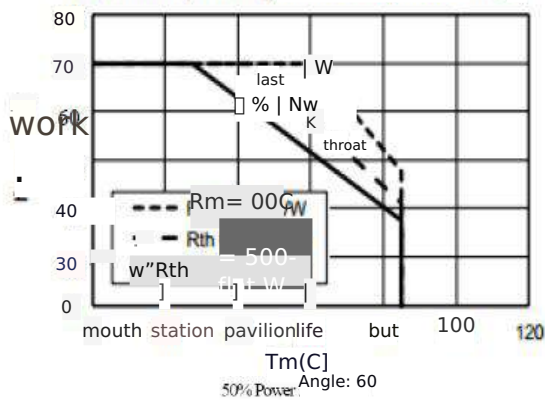
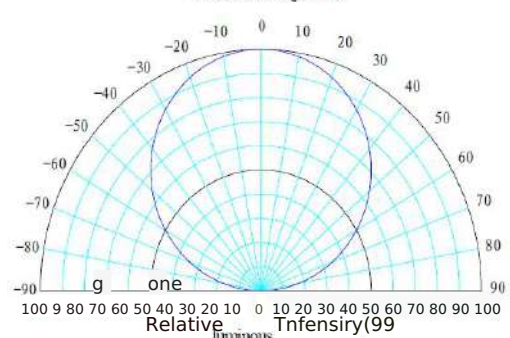
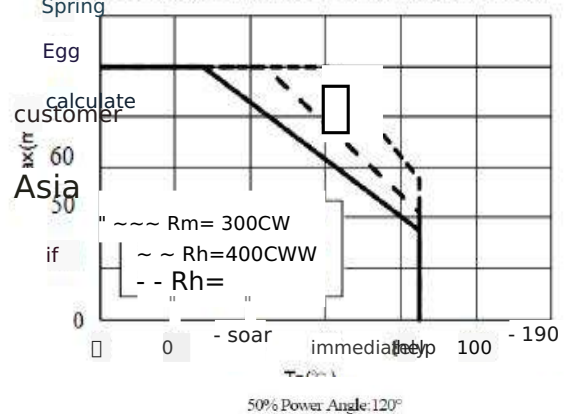


Fig.6- Maximum Driving Forward DC Current vs Ambient Temperature (Derating based on $T_{j\max}=110^{\circ}\text{C}$)



7. LED bulb reliability test conditions and criteria

Test items	Standards	Test adjustment	Note	Acceptance Scope
Resistance to soldering Heat	JEITA ED-4701 300 302	Ta=250±5°C 5sec 3mm from the base of the epoxy bulb	1 time	0/22
Solder ability	JEITA ED-4701 300 303	Ta=235±5°C 5sec	1 time over 95%	0/22
Mechanical Strength of Terminal (bent Test)	JEITA ED-4701 400 401	Load 5N(0.5kgf) 0°~90°~0° bond 3 times	No noticeable damage	0/22
Mechanical Strength of Terminal (pull Test)	JEITA ED-4701 400 401	Load 10N (1kgf) 10±1sec	No noticeable damage	0/22
Heat shock	JEITA ED-4701 300 307	0°~100° 15min 15min	100cycles	0/22
Temperature cycle	JEITA ED-4701 100 105	-40°C~25°C~100°C~25°C 30min 5min 30min 5min	100 cycles	0/22
Steady State Operating Life		Ta=60° RH=85% IF=80mA	500hrs (168,500)	0/22
High Temperature Storage	JEITA ED-4701 200 201	Ta=100°C	1000hrs (168,500,1000)	0/22
Temperature Humidity Storage	JEITA ED-4701 200 202	Ta=60°C RH=85%	1000hrs (168,500,1000)	0/22
Low Temperature Storage	JEITA ED-4701 200 202	Ta=-40°C	1000hrs (168,500,1000)	0/22
Life Test		Ta=25°C IF=80mA	1000hrs (168,500,1000)	0/22
Low Temperature Life Test		Ta=-40°C IF=80mA	500hrs (168,500)	0/22

8. Lamp Board Reliability Test Conditions and Acceptance Criteria

Test Items	Reference Standards	Test Conditions	Test Quantity	Receiving Range
Individual Vibration	GB/T 2423.10-2008	8-10 Hz 1.5 mm x, y, z axes in three directions, each 30 min	5 pcs	0/5
Post-packaging Vibration	GB/T 4857.23-2003	4-8 Hz 10 mm x, y, z directions, each 30 min	1 case	0/1
Drop Test	GB/T 4857.5-92	1 m / 1 time	5 pcs	0/5
Switch Illumination		Adjust Illumination Test On/Off Illumination	Full Measurement	0
Room Temperature Aging		25°C 100 h	5 pcs	0/5
High-Temperature Aging		60°C 100 h	5 pcs	0/5
Thermal Shock/Cycling		-35°C (30 min) ~ 60°C (30 min) 100 cycles	5 pcs	0/5

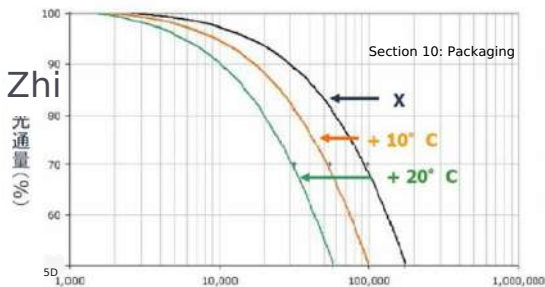
Section 9: Storage and Usage Precautions

◆ Storage

- The LED panels are large in size and are not sealed during shipment; it is best to use them immediately. If immediate use is not possible, store them in a dedicated electronics material storage warehouse.
- During storage, avoid stacking and heavy compression as much as possible.
- Should be stored in a dedicated electronics material warehouse. If not used for a long time, seal the outer packaging with cling film to prevent corrosion of component pins in high humidity.

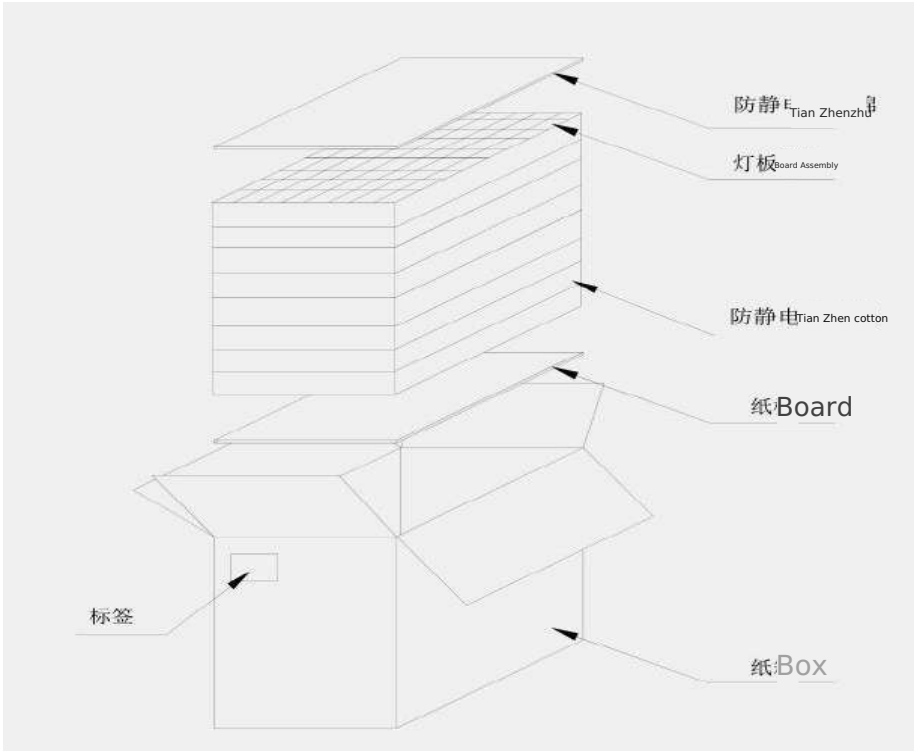
◆ Use

- Because LED panels contain LEDs and photosensitive devices, they are very temperature-sensitive, so during use the panel temperature should not exceed 85°C.
- Photosensitive devices may drift as temperature rises.
- The light-emitting devices and photosensitive devices are electrostatic devices; during installation and use, take anti-static precautions.
- LED devices' lifespan decreases as temperature rises. Normally, the LED bead temperature should not exceed 75°C; for every 10°C increase in temperature, its life is halved.



Tj (°C)	Lifespan (H)
X	97,000
X+10	55,000
X+20	32,000

Section 10: Packaging Method



If customers have special packaging requirements, package according to their specifications.